DO NOW — RECALL

# Boolean logic, gates and truth tables

Name: _____ Class: _____ Date: _____

**1. What is the difference between a stack and a queue when removing items?**

**2. What must be true about an array before you use binary search?**

**3. For int[] numbers = {5, 9, 14, 18, 23}; , what value is stored at zero-based index 3?**

**4. A binary search looks for 23 in the sorted array [2, 7, 11, 16, 23, 31, 44] . Which zero-based indexes are checked, in order?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A stack removes the most recently added item (LIFO); a queue removes the earliest added item (FIFO).

### 2. Model answer

The values must be sorted in the order expected by the search.

### 3. Model answer

18. C# arrays start at index 0.

### 4. Model answer

Indexes 3 → 5 → 4 (values 16 → 31 → 23). The search narrows the sorted range each time.

### By the end, you can…

• translate a problem into AND, OR, NOT and XOR logic

• convert between Boolean expressions and gate diagrams

• complete truth tables systematically

### Vocabulary — know these terms

Recall the term — Boolean

Recall the term — AND

Recall the term — OR

Recall the term — NOT

Recall the term — XOR

Recall the term — logic gate

Recall the term — truth table

Recall the term — expression

NEW LEARNING

# Read, annotate and explain

### From switches to logic gates

Imagine two switches, A and B , controlling a light, Q .

Learn what each gate does and looks like . Only then write expressions and combine gates.

Read the diagram from left (inputs) to right (output). A gate applies one rule.

### AND — both inputs must be 1

A gate takes an input (or inputs) and produces one output, Q . For AND, both inputs must be 1.

Example: 1 AND 0 = 0; 1 AND 1 = 1.

### OR — at least one input is 1

Either input can turn Q on. Both can be on too.

Example: 1 OR 0 = 1; 1 OR 1 = 1.

### NOT — reverse one input

NOT has just one input . The small circle on the output is the inversion symbol.

Example: NOT 1 = 0; NOT 0 = 1.

### XOR (⊻) — exactly one input is 1

XOR is different from OR: when both inputs are 1, XOR gives 0. Notice the extra curved line at the input.

Symbol: Q = A ⊻ B. Compare: 1 ∨ 1 = 1, but 1 ⊻ 1 = 0.

### Words first, then Boolean symbols

These symbols are shorthand for the rules you have just seen:

Write XOR using ⊻ : Q = A ⊻ B (1 only when A and B differ). Unlike OR (∨), XOR excludes the case where both inputs are 1. A, B and Q name inputs and output, not gates.

### Connect two gates in order

Say the logic aloud: A OR B, then AND C.

Brackets mean the OR gate is worked out before its output goes into AND.

### Check the circuit with a truth table

For three inputs there are 8 combinations . Work out the OR column first, then use that result with C.

Q = (A ∨ B) ∧ C. Only set Q to 1 when C is 1 and A∨B is 1.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

## Secure door access

### What switches ON?

```
Q = door unlocks
```

Start with the result the question is asking us to control.

### What are the alternatives?

```
A ∨ B
```

A card OR PIN works. OR (∨) means at least one; both can also be true. Do not write ⊻ here: XOR would wrongly stop access when both card and PIN are valid.

### What must also be true?

```
(A ∨ B) ∧ C
```

The building must be open , so join the options with AND.

### What must be OFF?

```
¬D
```

Lockdown must not be active. NOT reverses D.

### Join the chunks

```
Q = (A ∨ B) ∧ C ∧ ¬D
```

Work backwards: the final AND checks all requirements. Draw OR, AND and NOT first, then connect them to the final AND.

### Use intermediate columns

For Q = (A ∨ B) ∧ ¬C, three inputs give $2^3 = 8$ combinations. Write every input row, calculate A∨B, then ¬C, then combine those two columns with AND.

```
A B C | AvB | ¬C | Q
0 0 0 |  0  |  1 | 0
0 0 1 |  0  |  0 | 0
0 1 0 |  1  |  1 | 1
0 1 1 |  1  |  0 | 0
1 0 0 |  1  |  1 | 1
1 0 1 |  1  |  0 | 0
1 1 0 |  1  |  1 | 1
1 1 1 |  1  |  0 | 0
```

Example row 010: A∨B = 1 and ¬C = 1; therefore Q = 1. Always complete the intermediate columns before Q.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

### 1. Successful login

A login succeeds only when the user ID is valid AND the password is correct . Find the output, options, requirements and exceptions. Then write the expression and draw the gate.

```
A B | Q
0 0 | ?
0 1 | ?
1 0 | ?
1 1 | ?
```

Method / working:

### 2. Two test switches

A test lamp lights when exactly one of switches A and B is ON, but not both . Compare this with a different instruction: “at least one switch is on”.

```
A B | EXACTLY ONE | AT LEAST ONE
0 0 |      ?      |      ?
0 1 |      ?      |      ?
1 0 |      ?      |      ?
1 1 |      ?      |      ?
```

Method / working:

### 3. Library printer

The printer runs if a staff job OR a pupil job is waiting, but only if there is NOT a paper jam . Find the OR group, then the condition that prevents the output.

```
Q = (A ? B) ? ?C
```

Method / working:

### 4. Delivery bay light

A bay light comes on when movement is detected OR the door is open , but only when power is enabled and the maintenance lock is NOT active. Use all five steps. Decide which gate must feed Q last.

Method / working:

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

## Question 1

Burglar alarm circuit
A burglar alarm has four inputs. Each input is 1 when the condition is true and 0 otherwise:
A: door sensor detects movement
B: motion sensor detects movement
C: alarm is armed
D: test mode is enabled
The alarm sounds ( Q = 1 ) only when at least one sensor detects movement , the alarm is armed , and test mode is off .
Your tasks: (1) Write the Boolean expression for Q. (2) Draw the logic-gate circuit using AND, OR and NOT gates. (3) Label A, B, C, D and Q.

## Question 2

Expression and truth table
For P = ¬(A ∨ B) ⊻ C, complete all eight output values.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Create logic from words
A cooling fan runs when temperature is high OR load is high, but only when maintenance mode is NOT enabled. Write the expression and truth table.

### Question 4

Expression to gate plan
Convert Q = (A ⊻ B) ∧ ¬C into a gate-by-gate plan and complete its truth table.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A three-input Boolean problem has eight possible input combinations in its truth table.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.